

Modular architecture

How to split a program in sub-program?

modularization

- ↳ logic units
- ↳ substitution

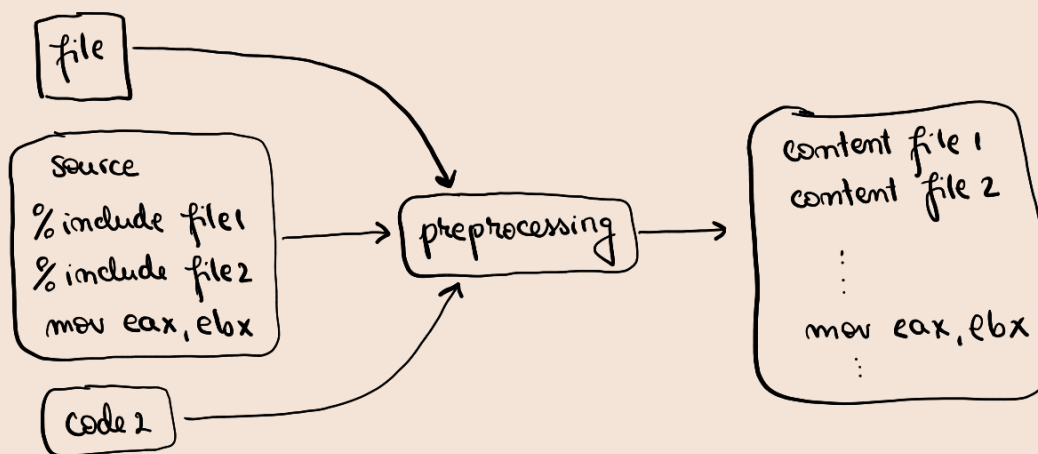
reusability

- ↳ files: % include
- ↳ binary files: libraries

techniques and tools

static inclusion at compile/assembly time: directive %include

- ↳ modularization: only code written in the same language (ex. asm)
- ↳ reusability: display the source code
- ↳ dangerous?
 - ↳ textual concatenation
 - ↳ exposes all variable names → conflicts
 - ↳ include all files



eg. %include

file: constants.inc

%ifndef -- CONSTANTS - INC -

%define -- CONSTANTS - INC -

MAX_BYTE equ 0xFF

MAX_WORD equ 0xFF

%endif

use constants.inc in an asm program

asm program

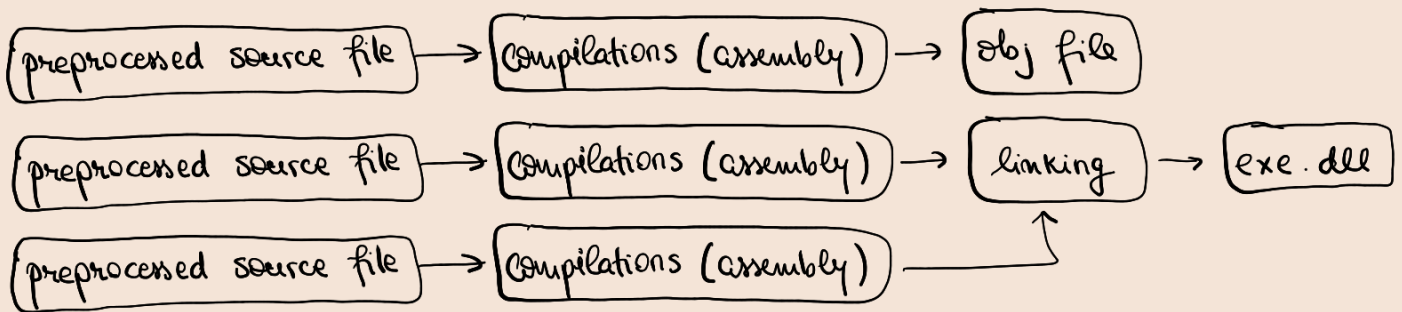
```

program.asm
%include constants.inc
; "moving" from EAX in a BYTE / WORD / DWORD
cmp eax, MAX-BYTE
je .no-fit-in-byte
    .fit-in-byte
    mov [result].al
    jmp .done
    :
    .done

```

Static linking resources (link-time)

↳ step performed at link-time (after assembly time)



- static linking : → text → text
(preprocessing)
 - assembler : instructions (text → binary encoding (object file))
 - compiler : instructions (text → binary encoding (object file))
 - linking : object file → library (program.dll / "exe")

• global and extern directives

• export through global

eg: global name1, name2

"name1" available to any interested file

• import through extern (request access to name3, name4.)

eg: extern name3, name4.

• request without availability ✗

• availability without request ✓

main.asm

sub.asm

- > `masm.exe - fobj main.asm → main.obj` } assembling the modules?
> `masm.exe - fobj sub.asm → sub.obj`
- edit links between the two modules
> `alink.exe main.obj sub.obj @PE-entry: start_subsys: console`] linking
 ↳ main.exe
> main.exe

Calling conventions : how to work with functions written in C.

CDECL (used at lab)

STDCALL, FASTCALL (on stack)

CDECL - C calling conventions

- ↳ pass param : stack (n params)
- ↳ volatile resources : EAX, ECX, EDI, EFLAGS
- ↳ return result : EAX, EDI:EAX
- ↳ cleanup : caller

STDCALL

- ↳ fix nr. params
- ↳ volatile resources : EAX, ECX, EDI, EFLAGS
- ↳ return result : EAX, EDI:EAX
- ↳ cleanup : caller

